Kaya-guided hybrid deep learning for leave-country-out estimation of national CO_2 emissions: a 112-country benchmark with country-clustered inference, conformal uncertainty, and explainable AI

Short title: Leave-country-out estimation of national CO_2 emissions

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Abstract

Many countries cannot yet produce complete and timely greenhouse-gas inventories, so their emissions have to be estimated from other information. We asked how accurately a country's per-capita fossil CO_2 emissions can be estimated from widely available energy and socio-economic indicators when none of that country's own data are used to fit the model, and whether deep learning improves on a simple linear model for this task. Using open data for 112 countries (1990–2023; 3,024 samples), we compared thirteen models, from ridge regression to a proposed hybrid network (KG-CBA) that adds a deep-learning correction to a linear component motivated by the Kaya identity. Every model was tested only on countries withheld from training (leave-

country-out cross-validation), and differences between models were tested with countries, not country-years, as the unit of inference. Ridge regression reached a median absolute percentage error (MdAPE) of 16.3% and was significantly more accurate than the tree-based models and the unguided convolutional networks (CNN-LSTM and CNN-BiLSTM-Attn). Guiding the network with the Kaya-motivated linear component reduced its MdAPE from 20.8% to 15.7% ($p = 0.002$), but no network or ensemble was shown to be more accurate than ridge regression: the best ensemble (15.1%) did not differ significantly from it ($p = 0.22$). Seven economies with unusual energy systems accounted for 78% of the squared error of ridge regression. Where the low-carbon share of primary energy was available (76 countries), adding this single indicator halved the MdAPE (15.6% to 7.7%, $p < 0.001$). The results were insensitive to the treatment of missing data and to the exclusion of the atypical economies. For estimating national emissions without a country's own records, a transparent linear model with the right energy indicators is sufficient, and better energy-mix statistics would improve the estimates more than more complex models.

Introduction

Carbon dioxide (CO_2) from fossil fuels and industry is the main driver of global warming, and emissions continue to set records [1,2]. Under the Enhanced Transparency Framework of the Paris Agreement, all Parties report national greenhouse-gas inventories [3], but the capacity to do so is uneven. Official inventories from many developing countries are infrequent or incomplete, and the gaps are filled with third-party estimates [4,5]. For fossil CO_2, independent data sets agree closely in aggregate and less closely country by country: global totals differ by about 5% once system boundaries are harmonized, whereas estimates for individual countries, regions or fuels can differ by 15% or more [6]. In Africa, two widely used data sets differ by less than 5%

for most of the main emitters, but by more than 10% for South Africa and by a factor of three for Mali, and African countries do not report every year [7]. Estimates that need only widely available indicators, and not a country's own emission accounts, are therefore useful for filling gaps and for cross-checking reported values.

Machine learning is increasingly proposed for this purpose. Comparative studies report that non-linear learners match or outperform classical time-series models for CO_2 series [8–10], with applications to transport [11,12], national totals [13] and sub-Saharan African economies [14]. Much recent work proposes hybrid architectures that stack convolutional, recurrent and attention layers and tunes them with metaheuristic search such as particle swarm optimization (PSO) [15,16].

Four features of this literature limit its value for the inventory-gap problem. First, most studies forecast future emissions of countries whose emission history is observed [8–10,13], the setting in which inventories already exist. Second, when the same country contributes data to both training and testing, a model can reproduce country-specific levels without having learned a transferable relationship; for grouped data the validation split must respect the groups [17,18]. Third, annual per-capita emissions change slowly, so a naive forecast that repeats the last observed value is already accurate, and accuracy that is not compared with it overstates skill [19,20]. Fourth, significance is usually assessed as if every country-year were an independent observation, although repeated years of one country are strongly correlated; ignoring this understates standard errors, often severely [21–23].

This study addresses these points by treating national emission estimation as a leave-country-out (LCO) problem: a country's per-capita CO_2 emissions are estimated from eight-year series of its energy and socio-economic indicators by a model fitted only to other countries. (We write LCO

because LOCO is established for leave-one-covariate-out analysis of variable importance [24], which we also use.) Because the target country contributes nothing to model fitting, no persistence forecast can be formed for it, and the task measures how well the relationship between indicators and emissions transfers between countries.

The task has a known structure. The Kaya identity expresses emissions as the product of population, income, the energy intensity of income and the carbon intensity of energy [25,26]; in logarithms it is additive, which is the basis of log-linear emission models [27]. A linear model in logarithms is therefore a natural reference, and the question for more flexible models is whether they add anything to it. Following the idea of theory-guided machine learning [28–30], we test this with a hybrid network, KG-CBA (Kaya-guided convolutional neural network, bidirectional long short-term memory and attention), in which a linear component carries the Kaya structure and a deep network models only what the linear component leaves unexplained.

The study answers four questions:

- Q1. How accurately can per-capita fossil CO_2 emissions be estimated for a country that is absent from training?
- Q2. Do deep and hybrid networks outperform a linear model motivated by the Kaya identity, once differences are tested with countries as the unit of inference?
- Q3. Is the remaining error reduced more by better models or by better covariate data?
- Q4. Are the answers robust to the treatment of missing data and to atypical economies, and how reliable are the prediction intervals and explanations of the models?

To our knowledge, national CO_2 models have not previously been compared under country-disjoint validation combined with country-clustered inference. A supporting experiment on

conventional temporal forecasting, in which naive persistence can serve as a benchmark, is reported in S2 Text.

Materials and methods

Study design and workflow

The study follows the seven steps of Fig 1. Three experiments use one data set and one validation design. Experiment A, the main benchmark, compares thirteen models on 112 countries with six covariates and answers Q1 and Q2. Experiment C repeats the comparison for the leading models on the 76 countries for which a seventh covariate, the low-carbon share of primary energy, is available, and answers Q3. Experiment B, a supporting analysis of conventional temporal forecasting in which naive persistence is a valid benchmark, concerns a different task and is given in S2 Text. The labels follow the order in which the experiments were run. Sensitivity, uncertainty and explainability analyses answer Q4.

Fig 1. Workflow of the study. The research questions (step 1) determine the data (step 2) and the three experiments (step 3). Thirteen models and an exploratory ensemble (step 4) are compared under leave-country-out cross-validation with country-clustered inference (step 5) and examined by diagnostic and robustness analyses (step 6), leading to the conclusions on models, data and practice (step 7). APE, absolute percentage error; CNN, convolutional neural network; GRU, gated recurrent unit; kNN, k-nearest neighbours; LSTM, long short-term memory; MAE, mean absolute error; MAPE, mean absolute percentage error; ML, machine learning; PSO, particle swarm optimization; RMSE, root-mean-square error; SHAP, Shapley additive explanations; SVR, support vector regression.

Data

All series were taken from the open CO_2 and energy data sets compiled by Our World in Data [31] (files `owid-co2-data.csv`, repository version of 2 June 2026, and `owid-energy-data.csv`, version of 27 April 2026; accessed on 26 July 2026). These files redistribute national fossil CO_2 emissions from the Global Carbon Budget 2025 [2], primary energy consumption from the Energy Institute Statistical Review of World Energy [32] and the U.S. Energy Information Administration [33], gross domestic product (GDP) from the Maddison Project Database 2023 [34], and population estimates.

The target is annual per-capita CO_2 emissions from fossil fuels and industry, excluding land-use change (c, in t CO_2 per person). Experiment A uses six covariates: primary energy consumption per capita (E, kWh per person), GDP per capita (G, international dollars per person at 2011 prices), population (P, persons), and the shares of national fossil CO_2 emissions that come from coal, oil and gas (s^{coal}, s^{oil} and s^{gas}; fractions between 0 and 1). Experiment C adds the share of primary energy supplied by low-carbon sources, that is, renewables and nuclear power (s^{lc}; fraction between 0 and 1), which the Statistical Review provides for 76 of the 112 countries.

We retained the years 1990 to 2023 and excluded regional aggregates. Missing values were filled in three steps for each series of each country: gaps were linearly interpolated for at most three consecutive years; values still missing were set to the last available value before them; and values missing at the start of a series were set to its first observed value. A country was retained if all seven series were then complete for all 34 years. The result is a balanced panel of 112 countries and 34 years, or 3,808 country-years (S1 Table lists the countries). Table 1 summarizes the variables, and Fig 2 shows the target and every covariate over the study period.

Table 1. Variables of the panel (112 countries, 1990–2023, 3,808 country-years).

Variable (symbol)	Unit	Source	Median (interquartile range)	Filled country-years, n (%)	Longest filled run, years
Per-capita CO_2 emissions (c), target	t CO_2 per person per year	[2]	4.45 (1.67–8.36)	0 (0.0)	0
Primary energy per capita (E)	kWh per person	[32,33]	22,121 (8,116–43,035)	28 (0.7)	16
GDP per capita (G)	International dollars per person, 2011 prices	[34]	12,688 (5,643–27,786)	114 (3.0)	2
Population (P)	Million persons	[31]	11.2 (5.1–41.7)	0 (0.0)	0
Coal share of CO_2 (s^{coal})	Fraction	[2]	0.12 (0.01–0.35)	244 (6.4)	26
Oil share of CO_2 (s^{oil})	Fraction	[2]	0.50 (0.33–0.66)	0 (0.0)	0
Gas share of CO_2 (s^{gas})	Fraction	[2]	0.18 (0.05–0.33)	291 (7.6)	31
Low-carbon share of primary energy (s^{lc}), 76 countries	Fraction	[32]	0.11 (0.03–0.22)	0 (0.0)	0

"Filled" counts values that were missing in the source and were completed as described in the text. The longest filled run is the largest number of consecutive filled years in any one country.

Fig 2. The target variable and the covariates, 1990–2023. Lines show the median across countries, dark bands the 25th to 75th percentiles and light bands the 10th to 90th percentiles. (a) Per-capita CO_2 emissions. (b) Primary energy per capita. (c) Gross domestic product (GDP) per capita in international dollars at 2011 prices. (d) Population. (e–g) Shares of fossil CO_2 emissions from coal, oil and gas. (h) Low-carbon share of primary energy. Panels a–g show the 112 countries of Experiment A and panel h the 76 countries of Experiment C. Panels a–d have logarithmic vertical axes.

The amount of filling differs between variables (Table 1). The target, population and the oil share needed none. Of the 114 filled values of GDP per capita, 112 are the year 2023, which the Maddison database does not yet cover, so that the 2022 value was carried forward. The coal and gas shares were filled for 244 and 291 country-years, and here the filled runs are long: 25

149 countries have a filled run of more than three years and 19 a run of more than ten years (S1
150 Table).
151 These long runs do not stand for unknown quantities. In the source, the emissions of a fuel are
152 blank where that fuel does not appear in the national total: in all 464 country-years with a blank
153 coal or gas entry, the reported total equals the sum of the reported components (coal, oil, gas,
154 cement, flaring and other industry) to within 0.3% of the total. The blank shares are therefore
155 zeros, or nearly so, for fuels that a country did not burn in those years, and the fill rule replaced
156 them with small positive values: 81% of the filled coal shares and 68% of the filled gas shares
157 are below 0.01, and 95% and 90% are below 0.05. We kept the original fill in the main analysis
158 so that all models, including the neural networks, were trained on the same panel, and we
159 examine alternatives in the sensitivity analyses.
160 Per-capita emissions range from 0.03 to 364.8 t CO_2 per person. The largest value (Kuwait,
161 1991, the year of the oil-well fires) precedes the first target year; the largest modelled value is
162 76.7 t per person (Qatar, 1997).

163 # Estimation task and validation

164 Let i index countries and t years. The three strictly positive covariates and the target are log-
165 transformed (Eq 1):

166 $$e_{i,t} = \log_{10} E_{i,t}, \qquad g_{i,t} = \log_{10} G_{i,t}, \qquad p_{i,t} = \log_{10} P_{i,t}, \qquad y_{i,t} = \ln c_{i,t}. \qquad (1)$$

167 Each sample pairs the covariates of $W = 8$ consecutive years with the target in the last of those
168 years (Eq 2):

169 $$\mathbf{X}_{i,t} = \left[\mathbf{x}_{i,t-W+1}, \ldots, \mathbf{x}_{i,t}\right], \qquad \mathbf{x}_{i,\tau} = \left(e_{i,\tau}, g_{i,\tau}, p_{i,\tau}, s_{i,\tau}^{coal}, s_{i,\tau}^{oil}, s_{i,\tau}^{gas}\right), \qquad (2)$$

where $\mathbf{x}_{i,\tau}$ is the vector of the $F = 6$ covariates of country i in year τ ($F = 7$ in Experiment C, which appends $s_{i,\tau}^{lc}$) and $\mathbf{X}_{i,t}$ is the window of $W \times F$ values. With 34 years per country this gives 27 samples per country, with target years 1997 to 2023: 3,024 samples for the 112 countries of Experiment A and 2,052 for the 76 countries of Experiment C. A model is a function f that returns the estimate $\hat{y}_{i,t} = f(\mathbf{X}_{i,t})$; the estimate of per-capita emissions is $\hat{c}_{i,t} = \exp(\hat{y}_{i,t})$. Models were evaluated by leave-country-out cross-validation. The countries were divided into five folds of 22 or 23 countries, with all years of a country in the same fold; each fold was held out in turn, and the model was fitted to the countries of the other four. Every result is therefore for countries none of whose data were used to fit the model that produced the estimate. Within each training set, one-seventh of the countries (rounded down: 12 countries in Experiment A, 8 in Experiment C) formed an inner validation set for early stopping, hyperparameter search and the calibration of prediction intervals. In each fold the covariates were standardized (Eq 3):

$$\tilde{x} = \frac{x - \mu_{tr}}{\sigma_{tr}}, \qquad (3)$$

where μ_{tr} and σ_{tr} are the mean and the standard deviation of the covariate over the training countries of that fold; $\tilde{\mathbf{x}}$ and $\widetilde{\mathbf{X}}$ denote the standardized covariate vector and window.

The fuel shares are ratios of a country's own fuel-specific emissions to its total emissions. They carry no information on the level of emissions, which is the quantity estimated, and no lagged value of the target enters any model. They would not, however, be available for a country without any emission accounts, where they would have to be derived from the national energy balance; the sensitivity analyses therefore include a model without them.

Models

Kaya identity and the linear reference model

The Kaya identity [25] writes national emissions as a product of four factors (Eq 4):

$$C = P \times \frac{Y}{P} \times \frac{Q}{Y} \times \frac{C}{Q}, \qquad (4)$$

where C is total fossil CO_2 emissions, P population, Y GDP and Q primary energy consumption. C appears on both sides because the expression is an identity; the factors are income per person (Y/P), the energy intensity of income (Q/Y) and the carbon intensity of energy (C/Q). Dividing by population and taking logarithms gives per-capita emissions $c = C/P$ in terms of energy per capita $E = Q/P$ (Eq 5):

$$\ln c = \ln E + \ln \frac{C}{Q}. \qquad (5)$$

Carbon intensity depends mainly on the fuel mix, so Eq 5 suggests that a model that is linear in the covariates of Eq 2 should explain most of the variation in y between countries. The linear reference model is ridge regression [35] on the standardized window (Eq 6):

$$\widehat{\boldsymbol{\beta}} = \underset{\boldsymbol{\beta},\, \beta_0}{\operatorname{argmin}} \sum_{n=1}^{N_{tr}} (y_n - \beta_0 - \boldsymbol{\beta}^{\top} \tilde{\mathbf{z}}_n)^2 + \lambda \parallel \boldsymbol{\beta} \parallel_2^2, \qquad (6)$$

where n indexes the N_{tr} training samples, $\tilde{\mathbf{z}}_n$ is the standardized window $\widetilde{\mathbf{X}}_n$ written as one vector of $W \times F = 48$ values, $\boldsymbol{\beta}$ and β_0 are the coefficients and the intercept, and $\lambda = 1$ is the penalty on the squared size of the coefficients.

Benchmark models

Thirteen models were compared (Table 2). They span the model families used in the emission-modelling literature, in increasing order of flexibility: a naive baseline; the linear reference

model; five classical machine-learning models (k-nearest neighbours, support vector regression [36], random forest [37], XGBoost [38] and LightGBM [39]); four deep sequence networks that are the building blocks of published hybrids (long short-term memory, LSTM [40]; gated recurrent unit, GRU [41]; a convolutional network followed by an LSTM, CNN-LSTM; and a convolutional network followed by a bidirectional LSTM with attention [42], CNN-BiLSTM-Attn); and the two Kaya-guided hybrids introduced below. The mean predictor is not a fitted model in the usual sense: it assigns to every test sample the mean of y over the training countries and shows the error obtained without any covariate information. All models used the same folds and inputs. The equations of the recurrent, convolutional and attention layers are standard and are given, with all symbols defined, in S1 Text.

Table 2. Models and their settings.

Model	Family	Settings
Mean predictor	Naive baseline	Mean of y over the training countries; no covariates
Ridge regression	Linear	Penalty λ = 1 (Eq 6)
k-nearest neighbours (kNN)	Classical machine learning	7 neighbours, distance-weighted
Support vector regression (SVR)	Classical machine learning	Radial basis kernel, C = 10, ε = 0.01
Random forest	Classical machine learning	600 trees, maximum depth 14
XGBoost	Classical machine learning	800 trees, learning rate 0.03, maximum depth 6, row subsample 0.85
LightGBM	Classical machine learning	800 trees, learning rate 0.03, 63 leaves (Experiment C: also row and column subsample 0.85, L2 penalty 1.5)
LSTM	Deep network	LSTM, 48 units
GRU	Deep network	GRU, 48 units
CNN-LSTM	Deep network	Two convolutional layers (48 filters, kernel size 3), then LSTM (48 units)
CNN-BiLSTM-Attn	Deep network	Two convolutional layers (48 filters, kernel size 3), bidirectional LSTM (48 units), additive attention
KG-CBA	Kaya-guided hybrid	Linear head started at the ridge solution; residual branch as CNN-BiLSTM-Attn with a 32-unit dense layer (Eq 7); fixed settings of the deep networks: 48 filters, kernel size 3, 48 LSTM units, dropout 0.25, learning rate 10^{-3}, batch size 96

Model	Family	Settings
PSO-KG-CBA	Kaya-guided hybrid	As KG-CBA, with settings selected by PSO: 37 filters, kernel size 4, 40 LSTM units, dropout 0.231, learning rate 9.96×10^{-4}, batch size 48
Ensemble	Exploratory	Mean of the estimates of ridge, GRU and PSO-KG-CBA

Classical models use the window as one vector of 48 values; deep networks use it as an 8×6 array. Deep networks were trained with the Adam optimizer on the mean squared error of y (dropout 0.25, learning rate 10^{-3}, batch size 96, at most 90 epochs), and training stopped when the error on the inner validation countries had not improved for 9 epochs. PSO search ranges: 16–64 filters; kernel size 2, 3 or 4; 16–96 LSTM units; dropout 0.05–0.45; learning rate 2×10^{-4} to 3×10^{-3}; batch size 48, 64, 96 or 128.

Kaya-guided hybrid network

KG-CBA writes the estimate as the sum of a linear term and a learned residual (Fig 3; Eq 7):

$$\hat{y} = \left(\boldsymbol{\beta}_K^{\top}\tilde{\mathbf{x}}_W + \beta_{K,0}\right) + r_{\boldsymbol{\theta}}(\tilde{\mathbf{X}}), \qquad (7)$$

where the term in parentheses is the Kaya head and $r_{\boldsymbol{\theta}}$ is the residual branch, a deep network with parameters $\boldsymbol{\theta}$; $\tilde{\mathbf{x}}_W$ is the standardized covariate vector of the most recent year of the window, and $\boldsymbol{\beta}_K$ and $\beta_{K,0}$ are the weights and intercept of the head. The weights of the head are started at the ridge solution for the most recent year, and the output weights of the residual branch are started close to zero. Training therefore begins from the ridge estimate, and the deep branch can only add what improves on it; the head and the branch are then trained together.

Fig 3. Architecture of the Kaya-guided hybrid network (KG-CBA). The standardized window of eight years and F covariates feeds two paths. The Kaya linear head acts on the most recent year and is started at the ridge solution. The deep residual branch (two one-dimensional convolutional layers, a bidirectional long short-term memory layer, attention pooling and a dense layer) models what the linear head does not explain. The two outputs are added to give the estimate of the logarithm of per-capita CO_2 emissions.

Six settings of KG-CBA were tuned by particle swarm optimization [16] with six particles and five iterations, that is, 30 evaluations (Table 2; S1 Text; S4 Fig); the budget was kept small because each evaluation trains a network. The fitness was the mean squared error on the inner validation countries of the first training fold. The selected configuration was used in all five folds, giving the model PSO-KG-CBA (E-PSO-KG-CBA in Experiment C). The untuned KG-CBA is reported next to it, so that the effect of tuning can be read directly from the results.

The ensemble averages the estimates of ridge, GRU and PSO-KG-CBA (ridge and E-PSO-KG-CBA in Experiment C). Its members were chosen after the out-of-fold results had been inspected. It is therefore an exploratory analysis whose accuracy is likely to be optimistic, and it was not a candidate for the reference estimator.

Model comparison and selection

Accuracy was measured on the original scale for the held-out samples. The absolute percentage error (APE) of a sample is $100\,|c - \hat{c}|/c$. We report its median (MdAPE) and mean (MAPE), the mean absolute error (MAE), the root-mean-square error (RMSE) and the coefficient of determination (R^2); the formulas are given in S1 Text. Per-capita emissions are strongly right-skewed (0.03 to 77 t per person among the targets), so RMSE and R^2 are dominated by a few high emitters, whereas MdAPE gives every country-year the same weight in relative terms. MdAPE is therefore the primary measure.

Each country contributes 27 consecutive, strongly correlated annual errors, so the 3,024 held-out errors are not independent observations. The validation folds respect this structure, because all years of a country fall in the same fold, and the statistical comparison of two models must respect it as well. For each pair of models we report (i) a Wilcoxon signed-rank test [43] on the 3,024 paired APE values, which treats country-years as independent and is shown only to make

the consequence of ignoring the clustering visible (the observation-level test); (ii) the same test on the 112 paired per-country medians of APE (the per-country test); and (iii) a cluster bootstrap [44] in which countries, with all their years, are resampled with replacement 10,000 times, giving a 95% percentile interval and a two-sided p-value for the median of the paired differences in APE. Conclusions are drawn from (iii), with (ii) as a check.

Models were ranked by MdAPE. The reference estimator was taken to be the simplest covariate-based model that no other model was shown to outperform significantly in the cluster bootstrap ($p < 0.05$).

Sensitivity analyses

Three sets of analyses test the robustness of the results. They were run for the ridge reference model and, where stated, for the five classical learners, all of which were re-estimated for this purpose; the neural networks were not re-trained.

Missing data. (i) The blank coal and gas shares were set to zero instead of being filled, as the accounting identity described under Data implies. (ii) The analysis was repeated without the 25 countries that have a filled run of more than three years (87 countries remain) and without the 19 countries that have a run of more than ten years (93 remain). (iii) The reference model was scored separately on countries with and without long filled runs. Multiple imputation for panel data [45] was not used, because the missing entries are not unknown values.

Atypical economies. Seven economies whose energy systems the covariates describe poorly were identified from the error analysis (Qatar, Kuwait, Saudi Arabia, Singapore, Luxembourg, Norway and Iceland). Accuracy was recomputed without them, both by scoring the fitted models on the remaining 105 countries and by re-estimating the models on those countries.

Covariate set. Ridge regression was re-estimated without the three fuel shares, with energy per capita alone, and with the covariates of the most recent year only.

Prediction intervals

Prediction intervals were constructed for the PSO-tuned hybrid of Experiment C (E-PSO-KG-CBA) in three steps (formulas in S1 Text). First, dropout was kept active at prediction time (Monte-Carlo dropout [46]), and 60 repeated predictions of the same sample gave a predictive mean and standard deviation of y. Second, the intervals were calibrated by split-conformal prediction [47,48]: on the inner validation countries of each fold, the absolute error of the predictive mean was divided by the predictive standard deviation, and the calibration factor $\hat{q}$ was set to the 90th percentile of these ratios, with the usual finite-sample correction. Third, the 90% interval for a test sample was the predictive mean plus and minus $\hat{q}$ standard deviations, transformed back to t CO_2 per person. Intervals were assessed by their coverage, the percentage of held-out observations inside their interval, which should equal the nominal 90%, and by their mean width.

Conformal intervals attain their nominal coverage when calibration and test observations are exchangeable. Here they come from different countries, so that exchangeability holds, at best, between countries and not between country-years [49–51]. To separate this effect from the properties of one model, we repeated the calibration for ridge regression in Experiment C with calibration sets of 8 training countries (the size used for the hybrid), of all 60 or 61 training countries by grouped cross-conformal prediction and, as a reference in which exchangeability holds, of samples drawn from the held-out countries themselves (S1 Text). The same analysis for Experiment A, with 12 calibration countries, is reported in the Results.

Explainability

Three measures of the importance of each covariate were used. The first is SHAP (Shapley additive explanations) [52], which attributes each estimate of a fitted model to its inputs: the SHAP value of an input is its average contribution to the estimate over all subsets of the other inputs, and the mean of its absolute value over samples measures the overall importance of the input (formula and symbols in S1 Text). SHAP values were computed exactly for the LightGBM model of Experiment C with the tree algorithm [53] and aggregated by covariate and by year of the window.

SHAP describes how a fitted model uses its inputs. It does not measure how much an input contributes to accuracy for unseen countries, and it can mislead when inputs are correlated [54]. Two further measures were therefore defined directly on held-out error and computed for ridge regression and LightGBM on the 76 countries of Experiment C. For permutation importance [37,55], the eight-year series of one covariate was shuffled between the test samples of each fold (30 repetitions) and the increase in MdAPE recorded. For drop-one-covariate importance, the leave-one-covariate-out analysis of Lei et al. [24], the model was re-estimated without the covariate under the same folds. For the hybrid, the attention weights show how much each year of the window contributes to the pooled representation.

Software, data and code

The benchmark was run in Python 3.12.3 with TensorFlow 2.21.0 [56], scikit-learn 1.8.0 [57], XGBoost 3.4.0, LightGBM 4.7.0 and SHAP 0.52.0, with random seed 42 and fold-specific offsets. The analyses added in revision were run in Python 3.13 with scikit-learn 1.9.1, XGBoost 3.4.1, LightGBM 4.7.0 and SHAP 0.54.1; they reproduce the original results for the mean predictor, ridge regression, k-nearest neighbours and support vector regression to three decimal

places, whereas the three tree ensembles were re-estimated with a different random state and differ slightly (S2 Table). The harmonized panel, the code for every model, the out-of-fold estimates of every model and the scripts that produce each table and figure are available at https://github.com/samfozzy/kg-cba-pclm-revision (archived release DOI: https://doi.org/10.5281/zenodo.24195216). The source data are available from https://github.com/owid/co2-data and https://github.com/owid/energy-data. The use of a generative artificial intelligence tool in this work is described in the Acknowledgments.

Results

Data overview

Median per-capita emissions were 5.0 t CO_2 per person in 1990 and 4.0 t in 2023, and the spread between countries remained wide: in 2023 the 10th and 90th percentiles were 0.6 and 11.5 t per person (Fig 2a). Across all country-years, the logarithm of per-capita emissions correlates strongly with the logarithm of energy per capita (Pearson r = 0.97) and of GDP per capita (r = 0.88), and only weakly with the fuel shares (S2 Fig), as the Kaya identity leads one to expect.

Experiment A: accuracy of the thirteen models

Table 3 and Fig 4 give the accuracy of every model on held-out countries (S1 Fig shows estimated against observed values). Covariates are clearly informative: the mean predictor has an MdAPE of 63.3%, and every covariate-based model except k-nearest neighbours (39.0%) is below 22%.

Ridge regression has an MdAPE of 16.3%. It has the lowest MAE and RMSE and the highest R^2 of all models, and a lower MdAPE than every classical machine-learning model and every unguided deep network (18.2% to 39.0%).

Kaya guidance changes the deep network substantially. The unguided CNN-BiLSTM-Attn network has an MdAPE of 20.8%; with the linear head, the same architecture reaches 15.7% (KG-CBA) and 16.7% (PSO-KG-CBA). The hybrids nevertheless have a higher RMSE (4.6 to 4.7 t per person) and a lower R^2 (0.52 to 0.55) than any other deep network, and tuning by PSO did not improve on the untuned hybrid. The exploratory ensemble has the lowest MdAPE (15.1%) and MAPE (23.5%).

Table 3. Accuracy on held-out countries in Experiment A (112 countries, six covariates, 3,024 samples).

Model	Family	RMSE	MAE	MAPE (%)	MdAPE (%)	R^2
Mean predictor	Naive baseline	7.415	4.408	337.8	63.3	−0.178
Ridge regression	Linear	**3.056**	**1.443**	25.2	16.3	**0.800**
kNN	Classical machine learning	5.119	2.729	49.7	39.0	0.439
SVR	Classical machine learning	4.881	2.100	32.3	21.3	0.490
Random forest	Classical machine learning	4.233	1.824	28.7	19.1	0.616
XGBoost	Classical machine learning	4.043	1.775	28.0	18.4	0.650
LightGBM	Classical machine learning	3.943	1.813	28.4	18.6	0.667
LSTM	Deep network	3.815	1.703	27.6	18.2	0.688
GRU	Deep network	3.629	1.644	25.1	18.5	0.718
CNN-LSTM	Deep network	3.773	1.707	28.1	20.8	0.695
CNN-BiLSTM-Attn	Deep network	3.812	1.790	28.6	20.8	0.689
KG-CBA	Kaya-guided hybrid	4.595	1.898	26.0	**15.7**	0.548
PSO-KG-CBA	Kaya-guided hybrid	4.748	1.914	27.2	16.7	0.517
Ensemble (ridge, GRU, PSO-KG-CBA)	Exploratory	3.557	1.530	23.5	15.1	0.729

RMSE and MAE are in t CO_2 per person. Bold marks the best value among the thirteen models; the ensemble was assembled after the results had been inspected and is shown for exploration only. Abbreviations as in Table 2.

Fig 4. Accuracy of the models on held-out countries in Experiment A. (a) Median absolute percentage error. (b) Root-mean-square error. The dashed line marks ridge regression. Arrows indicate values beyond the axis range. Values are those of Table 3.

Country-clustered comparison of models

Table 4 gives the statistical comparisons. The effect of Kaya guidance is significant: KG-CBA and PSO-KG-CBA are more accurate than the unguided network of the same architecture (cluster-bootstrap p = 0.002 for each). Ridge regression is significantly more accurate than LightGBM (p = 0.006) and than the unguided CNN-BiLSTM-Attn network (p = 0.002). With the classical learners re-estimated for this revision, it is also significantly more accurate than XGBoost (p = 0.004), random forest (p = 0.011), support vector regression and k-nearest neighbours (both p < 0.001; S2 Table).

The ensemble is not significantly more accurate than ridge regression: the median paired difference in APE is −0.76 percentage points, with a 95% interval from −2.20 to +0.41 (cluster-bootstrap p = 0.22; per-country test p = 0.098). The observation-level test, which treats the 3,024 country-years as independent, gives $p = 7 \times 10^{-10}$ for the same comparison, eight orders of magnitude smaller. The ensemble is significantly more accurate than its two neural members, GRU (p < 0.001) and PSO-KG-CBA (p = 0.022).

KG-CBA has a slightly lower MdAPE than ridge regression (15.7% against 16.3%) and PSO-KG-CBA a slightly higher one (16.7%), while both have a markedly higher MAE and RMSE and a lower R^2 (Table 3). The cluster-bootstrap tests confirm that neither hybrid differs significantly from ridge regression: the median paired difference in APE is +0.48 points for KG-CBA (95% interval −0.95 to +1.98; p = 0.50) and +0.47 for PSO-KG-CBA (−1.05 to +2.09; p = 0.51). LSTM and GRU are not significantly less accurate than ridge regression either (p = 0.08

and 0.12), whereas the two convolutional networks are (p = 0.001 and 0.002). No model was thus shown to be more accurate than ridge regression, the simplest covariate-based model, and ridge regression is retained as the reference estimator.

Table 4. Paired comparisons of models with countries as the unit of inference.

Exp.	Comparison (first against second)	MdAPE (%), first / second	Median paired difference in APE, points [95% CI]	p, cluster bootstrap	p, per-country test	p, observation-level test
A	Ensemble against ridge	15.1 / 16.3	−0.76 [−2.20, +0.41]	0.22	0.098	6.9×10^{-10}
A	Ensemble against GRU	15.1 / 18.5	−2.57 [−3.77, −1.20]	0.0006	0.007	1.8×10^{-30}
A	Ensemble against PSO-KG-CBA	15.1 / 16.7	−1.33 [−2.85, −0.22]	0.022	0.074	3.6×10^{-31}
A	PSO-KG-CBA against CNN-BiLSTM-Attn	16.7 / 20.8	−3.51 [−5.97, −0.97]	0.0016	0.004	6.3×10^{-22}
A	KG-CBA against CNN-BiLSTM-Attn	15.7 / 20.8	−3.42 [−5.84, −1.21]	0.0018	0.004	4.9×10^{-26}
A	Ridge against LightGBM	16.3 / 18.6	−3.10 [−4.91, −1.03]	0.0058	0.026	8.3×10^{-25}
A	Ridge against CNN-BiLSTM-Attn	16.3 / 20.8	−4.29 [−7.05, −1.64]	0.0024	0.007	3.1×10^{-34}
A	KG-CBA against ridge	15.7 / 16.3	+0.48 [−0.95, +1.98]	0.50	0.58	1×10^{-4}
A	PSO-KG-CBA against ridge	16.7 / 16.3	+0.47 [−1.05, +2.09]	0.51	0.98	1×10^{-4}
A	LSTM against ridge	18.2 / 16.3	+1.55 [−0.18, +3.58]	0.079	0.17	$< 10^{-4}$
A	GRU against ridge	18.5 / 16.3	+1.69 [−0.44, +3.86]	0.12	0.18	$< 10^{-4}$
A	CNN-LSTM against ridge	20.8 / 16.3	+4.31 [+1.75, +6.81]	0.0012	0.006	$< 10^{-4}$
C	E-PSO-KG-CBA against ridge	8.09 / 7.72	+0.66 [−0.23, +1.57]	0.16	0.21	$< 10^{-4}$
C	Ensemble against ridge	7.35 / 7.72	−0.20 [−0.66, +0.36]	0.56	0.49	0.48
C	E-PSO-KG-CBA against LightGBM	8.09 / 13.10	−3.85 [−6.03, −1.95]	0.0001	0.0002	1.7×10^{-56}
C	Ensemble against E-PSO-KG-CBA	7.35 / 8.09	−0.86 [−1.45, −0.35]	0.0014	0.004	5.1×10^{-31}
C	Ridge with seven against ridge with six covariates	7.71 / 15.64	−7.07 [−9.61, −4.59]	< 0.0001	2.0×10^{-7}	6.9×10^{-121}

Exp., experiment. A negative difference means that the first model has the smaller error. The interval is the 95% percentile interval of a cluster bootstrap that resamples countries (10,000 draws). The per-country test is a Wilcoxon signed-rank test on per-country medians of APE (112

countries in Experiment A, 76 in Experiment C). The observation-level test is the same test on all country-years and is shown only to illustrate the effect of ignoring the clustering; conclusions follow the cluster bootstrap. The last row, and the rows against ridge regression in Experiment A, were added in revision and use identical countries and folds for both models; they were computed by a separate script with its own bootstrap draws, so their p-values can differ from the other rows by about 0.001. CI, confidence interval.

Error concentration in atypical energy systems

The squared error is concentrated in a few countries (Fig 5a). For ridge regression, the seven atypical economies account for 77.9% of the total squared error. For the exploratory ensemble and the PSO-tuned hybrid, the seven atypical economies account for 84.3% and 86.5% of the squared error (S3 Fig shows the ensemble). Qatar, Kuwait and Singapore are among the six worst-fitted countries under each of the six models examined; Iceland, Norway, Luxembourg and Saudi Arabia enter or leave the list depending on the model. Ridge regression overestimates the emissions of Iceland and Norway, whose energy is largely hydroelectric and geothermal (mean error +13.4 and +9.8 t per person), and of Singapore (+13.7 t per person), a refining and bunkering centre in which a large part of energy use is plausibly not emitted on its territory. It underestimates the emissions of the hydrocarbon exporters Qatar, Kuwait and Saudi Arabia (−4.6, −9.1 and −5.3 t per person) and of Luxembourg (−7.1 t per person).

Leaving these seven economies out changes the scale-dependent measures and little else (Table 5; Fig 5b). For ridge regression the RMSE falls from 3.06 to 1.48 t per person and R^2 rises from 0.80 to 0.90, while the MdAPE changes only from 16.3% to 15.1% (14.9% when the model is re-estimated without them). The five classical learners improve in the same way, and ridge regression remains the most accurate of the six on MdAPE (S2 Table). Without the seven

atypical economies, the RMSE is 1.45 t per person for the ensemble, 1.75 for KG-CBA and 1.80 for PSO-KG-CBA, against 1.48 for ridge regression, and the R^2 of the hybrids rises to 0.86 and 0.85 (ridge regression 0.90). The low R^2 of the hybrids in Table 3 is therefore produced mainly by a handful of countries, but removing them does not make the hybrids more accurate than ridge regression: their differences from ridge regression remain not significant (cluster-bootstrap p = 0.87 for KG-CBA, 0.89 for PSO-KG-CBA and 0.10 for the ensemble; S2 Table).

Fig 5. Concentration of the error of ridge regression and effect of excluding the atypical economies. (a) Share of the total squared error of ridge regression contributed by each of the twelve countries with the largest contributions. Orange bars are the seven atypical economies. Countries are identified by ISO 3166-1 alpha-3 codes: SGP, Singapore; ISL, Iceland; QAT, Qatar; NOR, Norway; KWT, Kuwait; LUX, Luxembourg; SWE, Sweden; SAU, Saudi Arabia; KAZ, Kazakhstan; MLT, Malta; ARE, United Arab Emirates; FRA, France. (b) Root-mean-square error of ridge regression and the five classical learners for all 112 countries and with the seven atypical economies excluded from scoring. The three tree ensembles were re-estimated for this analysis, and their values for all countries differ slightly from Table 3.

Table 5. Sensitivity of the ridge reference model to atypical economies, missing-data treatment and covariate set.

Analysis	N	RMSE	MAE	MAPE (%)	MdAPE (%)	R^2	Difference from reference, points [95% CI]; p
Reference: original panel, six covariates	112	3.06	1.44	25.2	16.3	0.800	–
Atypical economies							
Seven atypical economies excluded from scoring	105	1.48	0.90	22.1	15.1	0.897	–
Seven atypical economies excluded, model re-estimated	105	1.61	0.94	22.6	14.9	0.878	–
The seven atypical economies alone	7	10.79	9.59	71.9	40.3	0.428	–
Missing data							

Analysis	N	RMSE	MAE	MAPE (%)	MdAPE (%)	R^2	Difference from reference, points [95% CI]; p
Blank coal and gas shares set to zero	112	3.06	1.44	25.3	16.2	0.800	+0.03 [−0.02, +0.07]; 0.19
Countries with a filled run of more than 3 years excluded, model re-estimated	87	3.33	1.74	27.5	17.6	0.778	–
Countries with a filled run of more than 10 years excluded, model re-estimated	93	3.33	1.61	25.4	15.6	0.769	–
Reference model scored on the 87 countries without a long filled run	87	3.41	1.67	26.5	15.5	0.767	–
Reference model scored on the 25 countries with a long filled run	25	1.20	0.64	20.9	18.3	0.941	–
Covariate set							
Without the three fuel shares	112	3.24	1.57	26.7	16.9	0.775	+1.03 [−0.62, +3.06]; 0.22
Energy per capita only	112	3.60	1.69	28.5	17.9	0.723	+2.37 [+0.59, +4.17]; 0.009
Covariates of the most recent year only	112	3.34	1.52	25.4	15.7	0.761	+0.18 [−0.32, +0.72]; 0.41

All rows are leave-country-out results for ridge regression. N is the number of countries scored. RMSE and MAE are in t CO_2 per person. The last column gives the median paired difference in APE from the reference model with its 95% cluster-bootstrap interval and p-value, for analyses that use the same 112 countries and folds. A long filled run is a run of more than three consecutive filled years. Results for the five classical learners are given in S2 Table.

Experiment C: value of the low-carbon share

On the 76 countries for which the low-carbon share of primary energy is available, all four models are far more accurate than in Experiment A (Table 6). Ridge regression with seven covariates has an MdAPE of 7.7%, the hybrid 8.1%, the ensemble 7.35% and LightGBM 13.1%; ridge regression again has the lowest MAE, RMSE and MAPE and the highest R^2, and the ensemble is again not significantly more accurate than ridge regression (p = 0.56; Table 4). The gain is attributable to the covariate. With identical countries and folds, ridge regression has an MdAPE of 15.6% with six covariates and 7.7% with seven, a median paired difference of

−7.07 percentage points (95% interval −9.61 to −4.59; $p < 0.001$; Table 4). Fig 6 shows the corresponding comparison for ridge regression, the hybrid and the ensemble when the six-covariate estimates are taken from the models of Experiment A; the reduction is of similar size for all three.

The gain is not confined to the atypical economies. Among the 69 other countries of this panel, the MdAPE of ridge regression falls from 14.1% to 7.0% ($p < 0.001$). Among the atypical economies it falls sharply for the two hydroelectric and geothermal systems, from 157% to 29% for Iceland and from 107% to 28% for Norway, and moderately for Luxembourg, Saudi Arabia, Kuwait and Qatar. Singapore is the exception: its error increases from 125% to 231%. Its low-carbon share is close to zero, so the new covariate confirms a fossil-based energy system but cannot represent energy use that is not emitted on its territory. In total, the squared error falls by 39%, and Singapore then accounts for 49% of what remains.

Table 6. Accuracy on held-out countries in Experiment C (76 countries, 2,052 samples).

Model	Covariates	RMSE	MAE	MAPE (%)	MdAPE (%)	R^2
LightGBM	Seven	4.520	2.048	22.8	13.1	0.622
E-PSO-KG-CBA	Seven	3.682	1.593	16.0	8.1	0.749
Ridge regression	Seven	**3.037**	**1.284**	**14.1**	7.7	**0.829**
Ensemble (ridge, E-PSO-KG-CBA)	Seven	3.306	1.409	14.5	7.35	0.798
Ridge regression, same countries and folds	Six	3.877	2.003	22.5	15.6	0.722

RMSE and MAE are in t CO_2 per person. Bold marks the best value among the three individual models with seven covariates; the ensemble is exploratory. The last row was added in revision and differs from the six-covariate values shown in Fig 6, which come from the models of Experiment A (trained on 112 countries) scored on the same 76 countries (ridge regression: MAPE 22.2%, MdAPE 14.3%).

Fig 6. Effect of adding the low-carbon share of primary energy. (a) Mean absolute percentage error (MAPE) and (b) median absolute percentage error (MdAPE) on the same 76

countries for ridge regression, the particle-swarm-tuned hybrid (PSO-KG-CBA) and the exploratory ensemble. Grey bars: six covariates, estimates from the models of Experiment A. Green bars: seven covariates (Experiment C).

Robustness to missing-data treatment and covariate set

The treatment of the blank fuel shares does not affect the results (Table 5). Setting them to zero instead of filling them changes the MdAPE of ridge regression from 16.3% to 16.2% ($p = 0.19$) and that of each of the five classical learners by at most 0.3 percentage points (S2 Table). Excluding the countries with long filled runs and re-estimating the model gives an MdAPE of 17.6% when 25 countries are excluded and 15.6% when 19 are excluded. Scored separately, the 87 countries without a long filled run have an MdAPE of 15.5% and the 25 countries with one have 18.3%, while on MAPE the order is reversed (26.5% and 20.9%). Countries with long filled runs are thus neither consistently easier nor consistently harder to estimate.

The fuel shares contribute little. Without them, the MdAPE of ridge regression is 16.9%, not significantly different from the reference model ($p = 0.22$); with energy per capita alone it is 17.9% ($p = 0.009$). The eight-year window also contributes little to the median error: with the covariates of the most recent year only, the MdAPE is 15.7% ($p = 0.41$), although the RMSE is higher (3.34 against 3.06 t per person). The result does not depend on the ridge penalty: for λ between 0.01 and 100 the MdAPE lies between 16.0% and 16.4%.

Regional results and African countries

For ridge regression, the median APE by continent ranges from 13.6% in South America to 20.6% in North America, with Africa at 16.9% (S2 Table). By country, the median of the 112

per-country MdAPE values is 15.5% (interquartile range 9.6% to 23.8%); 70 countries are below 20% and 11 are above 50% (S1 Table).

Of the 17 African countries in the panel, eight have a per-country MdAPE below 15% (Cameroon, South Africa, Botswana, Nigeria, Tanzania, Senegal, Ghana and Egypt; 7.1% to 13.5%), and three are above 40% (Niger 44%, Mozambique 74% and the Democratic Republic of the Congo 149%). These three have very low emissions per person, so that small absolute errors are large in relative terms, and none of them is in the panel of Experiment C: of the 17 African countries, only Algeria, Egypt, Morocco and South Africa have the low-carbon share in the source used here. Fig 7 shows the observed emissions of South Africa and Egypt with the estimates of the Experiment C ensemble and 90% prediction intervals of the hybrid. The per-country MdAPE is 6.1% for South Africa and 3.4% for Egypt for the ensemble, 5.6% and 5.7% for the hybrid, and 5.5% and 4.9% for ridge regression.

Fig 7. Observed and estimated per-capita CO_2 emissions of South Africa (ZAF) and Egypt (EGY), with 90% prediction intervals. Estimates are those of the Experiment C ensemble (mean of ridge regression and E-PSO-KG-CBA), fitted without any data from the country shown, and the 90% split-conformal intervals are those of E-PSO-KG-CBA. Panel titles give the median absolute percentage error (MdAPE) for the country.

Prediction intervals

Uncalibrated Monte-Carlo dropout intervals of the hybrid of Experiment C are far too narrow: at a nominal level of 90% they cover 13.2% of the held-out observations, with a mean width of 0.47 t per person. After split-conformal calibration the coverage is 81.9% (81.0% to 83.0% across repeated Monte-Carlo runs) and the mean width 5.68 t per person; the calibration factor $\hat{q}$ varies between folds from 8.8 to 69.2.

The analysis for ridge regression shows where the remaining shortfall comes from (S2 Table). When calibration and evaluation samples come from the same held-out countries, coverage is 90.2%, as the theory requires. With calibration on 12 training countries it averages 87.4% (80.3% to 94.5% across random draws of countries) and ranges from 45% to 100% in single folds; with calibration on all 89 or 90 training countries it is 88.9% (73.8% to 96.5% by fold). Coverage is also uneven between countries: 68% of the countries have every year inside the interval, and 8% have half of their years or fewer inside it. The comparison for Experiment C, which is the setting of the hybrid's intervals, gives the following. With the 8 calibration countries used for the hybrid, ridge regression reaches 86.5% on average over 200 random draws of countries (77.2% to 96.5%), with a mean width of 5.14 t per person and coverage by fold from 52% to 100%. Calibration on the 60 or 61 training countries by grouped cross-conformal prediction gives 88.8% with a mean width of 3.76; 61% of the countries have every year inside the interval and 9% have half or fewer, against 57% and 17% for the stored intervals of the hybrid. The narrower intervals of ridge regression reflect its smaller errors in this experiment. Calibration on a random half of the held-out observations themselves gives 90.3%.

Covariate importance

SHAP values for LightGBM in Experiment C are dominated by energy per capita (mean absolute value 0.114), followed by the low-carbon share (0.016), the gas share (0.008) and population (0.007) (Fig 8a). By year of the window, the most recent year carries most of the attribution (0.118), the year before it much less (0.027), and each earlier year less than 0.008 (Fig 8b). The attention weights of the hybrid, in contrast, are close to uniform (0.095 to 0.142 against a uniform value of 0.125; S6 Fig).

The two measures based on held-out error give the same ranking for the first two covariates (Table 7). Removing energy per capita and re-estimating the model raises the MdAPE of ridge regression from 7.7% to 29.4% and that of LightGBM from 13.2% to 32.2%; removing the low-carbon share raises it by 7.9 and 3.2 percentage points; removing any other covariate changes it by about one percentage point or less. Permutation importance agrees for LightGBM. For ridge regression it assigns about 30 percentage points to each of the three fuel shares, although removing any one of them costs at most 1.1 points: the shares are strongly related to one another, so that shuffling one of them produces combinations that do not occur, from which the linear model extrapolates.

Fig 8. SHAP importance in the LightGBM model of Experiment C. Mean absolute SHAP value (a) by covariate and (b) by year of the eight-year window, where t-0 is the target year. log_epc, logarithm of energy per capita; log_gpc, logarithm of GDP per capita.

Table 7. Importance of each covariate in Experiment C by three measures (76 countries).

Covariate	SHAP rank, LightGBM	Permutation, ridge	Permutation, LightGBM	Drop-one, ridge	Drop-one, LightGBM
Energy per capita	1	+54.7	+43.6	+21.7	+19.0
Low-carbon share	2	+10.6	+5.8	+7.9	+3.2
Gas share	3	+31.8	+0.5	+1.1	−0.6
Population	4	0.0	−0.4	−0.1	−0.8
Oil share	5	+31.4	+0.2	+0.3	+0.1
Coal share	6	+30.0	+1.6	+0.4	+0.2
GDP per capita	7	+1.0	−0.4	−0.3	−0.7

Permutation and drop-one columns give the change in MdAPE on held-out countries, in percentage points, when the eight-year series of the covariate is shuffled between test samples (mean of 30 repetitions) or when the model is re-estimated without the covariate. The reference MdAPE is 7.7% for ridge regression and 13.2% for LightGBM; LightGBM was re-estimated for this analysis and differs slightly from Table 6 (13.1%). SHAP ranks are those of Fig 8.

Supporting experiment: temporal forecasting

When a country's own emission history is available, naive persistence is difficult to beat (S2 Text). For forecasts of 2021 to 2023 made one year ahead, only one learner improved significantly on persistence, by 7% of the mean absolute error; three and five years ahead the largest improvements were 9% and 19%. In a replication for 2015 to 2019, no learner improved significantly on persistence one or three years ahead and only one did so five years ahead, and the learner that performed best in one period was generally not the best in the other.

Discussion

Principal findings

The four questions have clear answers. (Q1) From six widely available indicators, per-capita fossil CO_2 emissions of a country that is absent from training can be estimated with a median error of about 16%, and of about 8% where the low-carbon share of primary energy is also known. (Q2) Deep and hybrid networks do not outperform a linear model motivated by the Kaya identity: tree ensembles and the unguided convolutional networks (CNN-LSTM and CNN-BiLSTM-Attn) are significantly less accurate than ridge regression, and neither the Kaya-guided network nor an ensemble was shown to be more accurate than ridge regression alone. (Q3) Better covariate data reduce the error far more than better models: no change of model lowered the median error by more than 1.2 percentage points relative to ridge regression, whereas one additional covariate halved it. (Q4) These results do not depend on how missing fuel shares are treated, on whether the atypical economies are included, or on the ridge penalty.

Why deep learning did not outperform the linear model

Five features of the task, each visible in the results, explain why additional model capacity did not help.

The relationship is close to linear by construction. By the Kaya identity, the logarithm of per-capita emissions is the sum of the logarithm of energy per capita and the logarithm of carbon intensity (Eq 5). Energy per capita alone correlates at r = 0.97 with the target and gives an MdAPE of 17.9% in a one-covariate ridge model. Little non-linear signal remains to be found.

The number of independent units is the number of countries. A model that must generalize to new countries learns from about 90 training countries, not from 2,400 training samples, because the years of one country are close to copies of one another: the covariates of the most recent year alone give the same median error as the eight-year window. Flexible models with thousands of parameters cannot be estimated reliably from 90 effectively independent units; they fit country-specific patterns that do not transfer. This agrees with evidence that tree ensembles and simple linear models remain competitive with deep networks on small tabular and time-series problems [20,58–60].

Held-out countries can lie outside the range of the training countries. The economies with the largest errors are the most extreme in energy use or emissions per person. Tree ensembles cannot produce estimates beyond the range of the training targets, and neural networks behave unpredictably outside their training range, whereas a linear model in logarithms extrapolates along the identity. The residual branch of the hybrids lowers the median error slightly but adds large errors for a few extreme economies, which is why their RMSE is the highest among the deep models.

The remaining error is caused by missing information, not by the functional form. What the linear model does not explain is the part of carbon intensity that the six covariates do not describe, such as non-fossil energy. No model can recover this from covariates that do not contain it; when one covariate that contains part of it was added, the error of ridge regression, the hybrid and the ensemble fell by a similar large amount (Fig 6).

Tuning added variance, not accuracy. The hybrid tuned by particle swarm optimization was no more accurate than the hybrid with fixed settings. A fitness value computed on 12 validation countries is noisy, and selecting the best of 30 configurations on such a value mostly selects noise; a larger search budget would not remove this limitation, which comes from the number of countries.

We therefore found no advantage of the complex models in accuracy, and they are less transparent and more costly to train. Their value in this study is methodological. They represent current practice and had to be tested under a valid protocol before they could be set aside, and the comparison between the guided and the unguided network shows that the deficit of deep networks in this task is a failure to learn the linear structure from few countries, not a lack of capacity; once this structure is supplied, nothing remains for the network to add. For inventory gap-filling we recommend ridge regression on logarithmic covariates.

The testing regime matters as much as the model. Treating country-years as independent would have declared the ensemble better than ridge regression with p below 10^{-9}, whereas the clustered test gives $p = 0.22$; studies that test at the level of the observation will regularly report small architectural gains as highly significant when they are not.

What limits accuracy: covariate data

Experiment C shows that the accuracy of the estimates is limited by the covariates. The low-carbon share of primary energy measures directly the part of carbon intensity that the fossil-fuel shares cannot represent, and after energy per capita it is the second most important covariate by SHAP, by drop-one importance for both models and by permutation importance for LightGBM (Table 7). The measures answer different questions and are consistent with one another. SHAP describes how much a fitted model relies on a covariate; energy per capita dominates because it carries the scale of emissions. Experiment C and the drop-one analysis measure what is lost in accuracy for unseen countries when a covariate is unavailable. Repeating that experiment for the covariate that SHAP ranks first shows the largest loss (the MdAPE of ridge regression rises from 7.7% to 29.4% without energy per capita), but energy per capita is available for every country. The low-carbon share is the most valuable covariate that many countries lack.

That lack is concentrated where estimates are most needed: in the source used here, the low-carbon share is available for 76 of the 112 countries of the panel and for only 4 of its 17 African countries, and the three African countries that are estimated worst are among those without it. By contrast, the long filled runs in the fuel-share series are not a sign of thin statistics: they are zeros for fuels that a country did not use, and their treatment has no effect on the results. The estimator also does not need the fuel shares, the only inputs derived from emission accounts; without them the median error is 16.9% instead of 16.3%. It can therefore be applied to a country that has energy and economic statistics but no emission inventory.

A roadmap for better national estimates

The results suggest the following priorities, in order of expected benefit.

1. *Extend energy-mix statistics.* Compiling the low-carbon share of primary energy, or the full breakdown of primary energy by source, from national energy balances for the 36 panel countries that lack it, and for countries outside the panel, is the step most likely to improve estimates; where it is available the median error is halved.
2. *Add covariates for the atypical systems.* The largest remaining errors point to variables that could be tested in the same way: international bunker fuel and refinery throughput for a trading hub such as Singapore, and flaring and the energy used in oil and gas processing for hydrocarbon exporters.
3. *Derive fuel shares from energy balances.* This would make the estimator fully independent of emission accounts, at a cost of less than one percentage point of median error.
4. *Keep the linear core and test additions against it.* Hierarchical models that share information between similar countries [61], and guided networks of the kind tested here, should be retained only if they are significantly more accurate than ridge regression under country-clustered inference.
5. *Adopt evaluation standards that match the task.* Studies of national emission modelling should validate on countries that are absent from training, use the country as the unit of inference, compare temporal forecasts with naive persistence, and report errors by country in addition to pooled measures such as R^2.
6. *Calibrate uncertainty on many countries.* Prediction intervals should be calibrated on as many countries as possible and their coverage reported by country; methods designed for grouped data [51] are the natural next step.

7. *Extend the scope.* The same design can be applied to sectoral and sub-national emissions and to other greenhouse gases, for which inventories are less complete than for fossil CO_2.

Limitations

First, the balanced panel contains 112 countries, and small and low-income countries, for which inventories are weakest, are under-represented; only 17 African countries are included. Second, the covariates are not independent of the statistical systems that produce inventories: primary energy statistics are themselves the main input to inventories, so the estimator is a cross-check and a means of filling gaps, not a substitute for national energy statistics. Third, the target is territorial fossil CO_2; emissions from land use and other gases are not covered. Fourth, the sensitivity analyses were run for ridge regression and the classical learners; the neural networks were not re-trained under the alternative treatments. Fifth, the hybrid was tuned with a small search budget on the first fold, and the selected configuration was used in the other folds, whose test countries had been part of the search data; as tuning did not improve accuracy, this cannot have inflated the results. Sixth, the ensemble is exploratory. Seventh, prediction intervals cover less than their nominal level for countries unlike those used for calibration. Finally, the results rest on one vintage of the source data and on annual national totals.

Conclusions

National per-capita fossil CO_2 emissions can be estimated for a country from its energy and economic indicators, without any of its own emission data, with a median error of about 16%, and of about 8% where the low-carbon share of primary energy is known. A ridge regression on logarithmic covariates, which follows from the Kaya identity, is the estimator to use: tree

ensembles and the unguided convolutional networks are significantly less accurate, and neither a Kaya-guided hybrid network nor an ensemble was shown to improve on it once countries, not country-years, are the unit of inference. The error is concentrated in a few economies whose energy systems the indicators describe poorly, and one additional energy-mix indicator halves it. Estimates of national emissions where inventories are missing are therefore improved most by extending energy-mix statistics to more countries, not by increasing model complexity.

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Use of artificial intelligence tools. The authors used the generative AI tool Claude (Anthropic, San Francisco, CA, USA; Claude Fable 5.2 (API: claude-fable-5-2)) in three ways. (i) Code: Claude drafted and revised Python code for data harmonization, model training, statistical tests and figures, following analysis specifications written by the authors. (ii) Analyses added in revision: the scripts for the sensitivity, importance and conformal analyses were drafted and first run with Claude from the public source files. (iii) Text: Claude assisted with drafting and language editing of the manuscript and of the response to the reviewers. No AI tool generated or altered data; all data come from the cited public sources. To evaluate the validity of the outputs, All authors have confirmed. the authors read and executed all code and regenerated every table and figure from the released scripts; confirmed that independently re-implemented deterministic models reproduce the reported values to three decimal places; checked every numerical statement against the script output and every reference against the publisher's record; and reviewed and edited all text. The authors take full responsibility for the content of this article.

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Supporting information

S1 Text. Supplementary methods. Notation; equations of the recurrent, convolutional and attention layers with all symbols defined; particle swarm update; algorithms for the particle

swarm search, the training of the hybrid network and the country-clustered comparison of two models; per-fold results.

S2 Text. Experiment B: temporal forecasting against naive persistence. Protocol, results for two evaluation periods (Table A) and scaled errors by horizon (Fig A).

S1 Table. Countries and data audit. For each of the 112 countries: continent, membership of the 76-country panel, number of filled country-years and longest filled run for each variable, and per-country median absolute percentage error of ridge regression. (XLSX)

S2 Table. Sensitivity analyses and additional results. Results of the missing-data and atypical-economy analyses for ridge regression and the five classical learners; paired tests of ridge regression against the re-estimated classical learners; results by continent; covariate importance; conformal coverage for ridge regression. (XLSX)

S1 Fig. Estimated against observed per-capita CO_2 emissions on held-out countries. Ridge regression, GRU, PSO-KG-CBA and the exploratory ensemble in Experiment A; both axes are logarithmic.

S2 Fig. Correlations between the target and the covariates. Pearson correlation coefficients between the log-transformed target and the six covariates over 3,808 country-years.

S3 Fig. Squared error by country for the exploratory ensemble in Experiment A. Bars show the sum of squared errors of the fifteen countries with the largest contributions (left axis) and the line the cumulative share of the total (right axis). Dark bars mark countries that are among the six worst-fitted under each of the six models examined, and light orange bars the other countries among the six worst-fitted of this model.

S4 Fig. Convergence of the particle swarm search. Validation mean squared error of each of the 30 evaluated configurations and the best value found so far.

S5 Fig. Training and inner-validation loss of the four deep networks with fixed settings in the first fold.

S6 Fig. Mean attention weight by year of the window for the hybrid network of Experiment C. The dotted line marks uniform weights.